Learning Mechanism for the Zeta Project

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Abstract

We present the learning mechanism of the Zeta project, a finite-state learner defined entirely over the field $\mathbb{F}_{13}$ and its quadratic extension $\mathbb{F}_{169}$. A stimulus is absorbed by an exact evolution rule; the response is refined by a correction map chosen from a cyclic family of 28 maps by minimising an integer coordinate mismatch; and every learned symbol is written as a further digit of a single 13-adic integer, the Witt history. Experience is stored in an integer ledger of capacity $13^3 = 2197$ and retrieved by a three-stage associative search ordered by 13-adic prefix depth, spectral flag and panel neighbourhood. We give complete definitions of all components and prove the structural properties that follow from them: bijectivity of the linear index, the exact selection rule, the decomposition of the state space into two chiral subspaces of dimension three, the 13-adic characterisation of prefix matching, the capacity bound of the ledger, the properties of the spectral invariant that keys the memory, and the Chinese-remainder bijection of the plaquette labels. All arithmetic is exact; no floating-point quantity occurs in the mechanism.

Keywords: finite fields, 13-adic integers, Witt history, associative memory, exact arithmetic.

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1 Introduction and Motivation

1.1 Motivation

Conventional learning systems represent their state by vectors of floating-point numbers and adapt it by numerical optimisation. Their behaviour therefore depends on rounding, on the order of evaluation and on the hardware, and two runs of the same system on the same data need not agree bit for bit. For systems that are meant to run for a long time, to be audited, and to be replayed exactly, these dependencies are a liability.

The Zeta project follows a different route. Its state space is a finite-dimensional vector space over the field $\mathbb{F}_{13}$ with 13 elements, its symbols are digits of 13-adic integers, and every operation is an integer computation. In such a setting a learning mechanism cannot be a gradient method. It must answer four elementary questions in exact terms: how a stimulus changes a state; which one of finitely many admissible corrections to apply; what to remember of the stimulus; and how to retrieve that memory later by a notion of similarity that is itself algebraic. This article gives the answers implemented in the learning module of the project [4], which builds on the exact arithmetic layer of the framework [3].

1.2 Overview of the mechanism

A stimulus is a pair of vectors in $\mathbb{F}_{13}^3$. The learner processes it in the following stages.

1. *Stimulus space* (Sect. 3). The $13^3 = 2197$ possible stimulus vectors are numbered by a base-13 linear index and compared by an integer coordinate mismatch.
2. *Evolution* (Sect. 4). A base state is coupled to the stimulus; the response splits into a sector fixed by the Tribonacci register and a residue.
3. *Golden orbit and selection* (Sects. 4 and 5). The response is refined by one of 28 correction maps indexed by a cyclic subgroup of $\mathbb{F}_{169}^{\times}$; the map is selected by minimal coordinate mismatch, ties being resolved by memory.
4. *Relaxation and chirality* (Sect. 6). Quadratic rescaling, projection to the fixed sector, and a decomposition into two chiral halves.
5. *Witt history* (Sect. 7). Every learned symbol becomes a higher digit of a single 13-adic integer.
6. *Ledger and retrieval* (Sect. 8). Experience is stored in an integer ledger of capacity 2197 and retrieved by a three-stage associative search.
7. *Stimulus index* (Sect. 9). The responses to all 2197 elementary stimuli are tabulated once and used for lookup.

1.3 Contributions

- A complete formal definition of the learner as a finite state machine whose state (h, r, k, L) consists of a digit string, a residue vector, a small integer, and a bounded list of integer records.
- Proofs of the structural properties that follow from the definitions: bijectivity of the linear index; the exact selection rule of the orbit index; the decomposition of the state space into two chiral subspaces of dimension three; the 13-adic characterisation of prefix matching; the capacity bound of the ledger; the properties of the spectral invariant τ that underlies the memory flags; and the Chinese-remainder bijection of the plaquette labels.
- An explicit description of the cyclic group of order 28 that indexes the correction maps, as the subgroup generated by $\Phi = 7 + 8\theta$ of $\mathbb{F}_{169}^{\times}$.
- Exact cost bounds, nine figures generated by exact integer computation, and a self-test suite (Appendix A).

1.4 Conventions and organisation

Elements of $\mathbb{F}_{13}$ are identified with their integer representatives $0, \dots, 12$; all counters are integers and are never reduced modulo 13; no floating-point quantity occurs anywhere in the mechanism. Names in typewriter font refer to the corresponding components of the implementation.

Section 2 fixes notation and the register on which the learner operates. Sections 3 to 6 treat the stimulus space, the evolution rule, the golden orbit, the selection of a correction, and the relaxations. Sections 7 and 8 describe the Witt history and the ledger; Section 9 the stimulus index; Section 10 finiteness and cost. Section 11 concludes, and Appendix A lists the self-test properties.

2 Preliminaries

2.1 Finite fields

Let $\mathbb{F}_{13}$ be the field with 13 elements (for background on finite fields see [1]) and let $\bar{x} \in \{0, \dots, 12\}$ denote the integer representative of $x \in \mathbb{F}_{13}$. The nonzero squares of $\mathbb{F}_{13}$ are $1, 3, 4, 9, 10, 12$; since 2 is not among them, $\theta^2 - 2$ is irreducible over $\mathbb{F}_{13}$ and

$$\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]/(\theta^2 - 2) \tag{1}$$

is the field with 169 elements. Its nontrivial automorphism is the Frobenius map $\sigma(a + b\theta) = a - b\theta = (a + b\theta)^{13}$. The norm $N(x) = x\,\sigma(x) = x^{14}$ takes values in $\mathbb{F}_{13}$, and the elements of norm one form the subgroup of order 14 of the cyclic group $\mathbb{F}_{169}^{\times}$ of order 168.

2.2 The Tribonacci register

The learner operates on the register defined by the companion matrix of the Tribonacci polynomial $x^3 - x^2 - x - 1$, reduced modulo 13:

$$T_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in M_3(\mathbb{F}_{13}). \tag{2}$$

Over $\mathbb{F}_{13}$ its characteristic polynomial factors as $x^3 - x^2 - x - 1 = (x - 7)(x^2 + 6x + 2)$, where the quadratic factor is irreducible with roots $10 \pm 7\theta \in \mathbb{F}_{169}$. Hence T_3 has the three distinct eigenvalues

$$\lambda_1 = 7, \qquad \lambda_2 = 10 + 7\theta, \qquad \lambda_3 = 10 - 7\theta, \tag{3}$$

with $\lambda_1 + \lambda_2 + \lambda_3 = 1$, $\lambda_2\lambda_3 = 2$, and $\lambda_1\lambda_2\lambda_3 = 1$. Define

$$\rho = \lambda_2/\lambda_3 = 8 + 5\theta, \qquad N(\rho) = 1. \tag{4}$$

Because the eigenvalues are distinct, T_3 is diagonalisable over $\mathbb{F}_{169}$, and the spectral projectors

$$Q_i = \prod_{j \neq i} \frac{T_3 - \lambda_j I}{\lambda_i - \lambda_j} \qquad (i = 1, 2, 3)$$

satisfy $Q_i Q_j = \delta_{ij} Q_i$, $Q_1 + Q_2 + Q_3 = I$, $T_3 = \sum_i \lambda_i Q_i$ and $\sigma(Q_2) = Q_3$ (entrywise conjugation). The matrices

$$P_1 = Q_1 = \begin{pmatrix} 1 & 7 & 10 \\ 3 & 8 & 4 \\ 7 & 10 & 5 \end{pmatrix}, \qquad P_{23} = Q_2 + Q_3 = I - P_1 \tag{5}$$

have entries in $\mathbb{F}_{13}$; P_1 has rank one. The vector $v_0 = (1, 3, 7)$ satisfies $T_3 v_0 = 7v_0$, hence $P_1 v_0 = v_0$ and $P_{23} v_0 = 0$. For $v \in V = \mathbb{F}_{13}^3$ the two *sectors* are

$$a(v) = P_1 v, \qquad r(v) = P_{23} v, \qquad a(v) + r(v) = v. \tag{6}$$

Finally let

$$G = 2P_1 - I = P_1 - P_{23}. \tag{7}$$

Then $G^2 = P_1 + P_{23} = I$, and $G = T_3^{84}$ (equivalently $\lambda_2^{84} = \lambda_3^{84} = -1$).

2.3 Fermion states and stimuli

A *fermion* is a pair $\Psi = (\psi^+, \psi^-) \in V \times V \cong \mathbb{F}_{13}^6$ (class `Fermion`). The *vacuum* is

$$\Psi_{\text{vac}} = (v_0, v_0). \tag{8}$$

A *stimulus* is a fermion $s = (s^+, s^-)$. The elementary stimuli used to tabulate the response of the learner have vanishing plus part, $s = (0, (a, b, c))$. Fermions are added componentwise, written $\oplus$. The coupling operator $\mathcal{D}_f(\,\cdot\,; D_W)$, parametrised by a matrix $D_W \in M_3(\mathbb{F}_{13})$, and the quadratic forms n_+, n_- (of the plus and minus components, respectively) are components of the framework; they are used but not redefined here.

2.4 State of the learner

Definition 2.1 (Learner state)**.** The state of a learner is a quadruple $\mathsf{S} = (h, r, k, L)$ where

- h is a Witt prefix, a finite string of base-13 digits (Sect. 7), initially empty;
- $r \in V$ is the last residue, initially 0;
- $k \in \{0, \ldots, 27\}$ is the last golden index (Sect. 4), initially 0;
- L is the experience ledger, a finite list of records (Sect. 8), initially empty.

2.5 Framework components

The learner uses the following components of the Zeta framework.

Symbol	Implementation	Role
$\mathcal{D}_f(\,\cdot\,; D_W)$	`Fermion.Df`	coupling of a state to D_W
γ_5	`Fermion.gamma5`	chirality operator (Sect. 6)
n_+, n_-	`nPlus`, `nMinus`	quadratic forms
phase_{168}, g_{00}, q_{top}	`phase168`, `g00`, `qTop`	phase in $\mathbb{Z}/168$, parity source, topological charge
B_k, $\gamma(j)$	`goldenB`, `goldenIndex`	correction maps and index reduction, `GOLDEN_ORDER`$= 28$
spectral flag	`spectralFlag`, `panelNeighbors`, `FLAG_COUNT`	discretisation of τ and panel neighbourhood (Sect. 8)
plaquette maps	`fermionToPlaquette`, `plaquetteToFermion`	bijection $\mathbb{Z}/168 \times \{0,1\} \leftrightarrow [0,56) \times [0,6)$
field operations	`modp`, `invp`, `safeDiv`, `isSquareF13`, `sqrtF13`	arithmetic in $\mathbb{F}_{13}$
Witt prefix	`WittPrefix`	13-adic digit strings

3 Stimulus Space and Linear Index

Definition 3.1 (Linear index)**.** For $(a, b, c) \in \mathbb{F}_{13}^3$,

$$\iota(a, b, c) = 169\,\bar{a} + 13\,\bar{b} + \bar{c}. \tag{9}$$

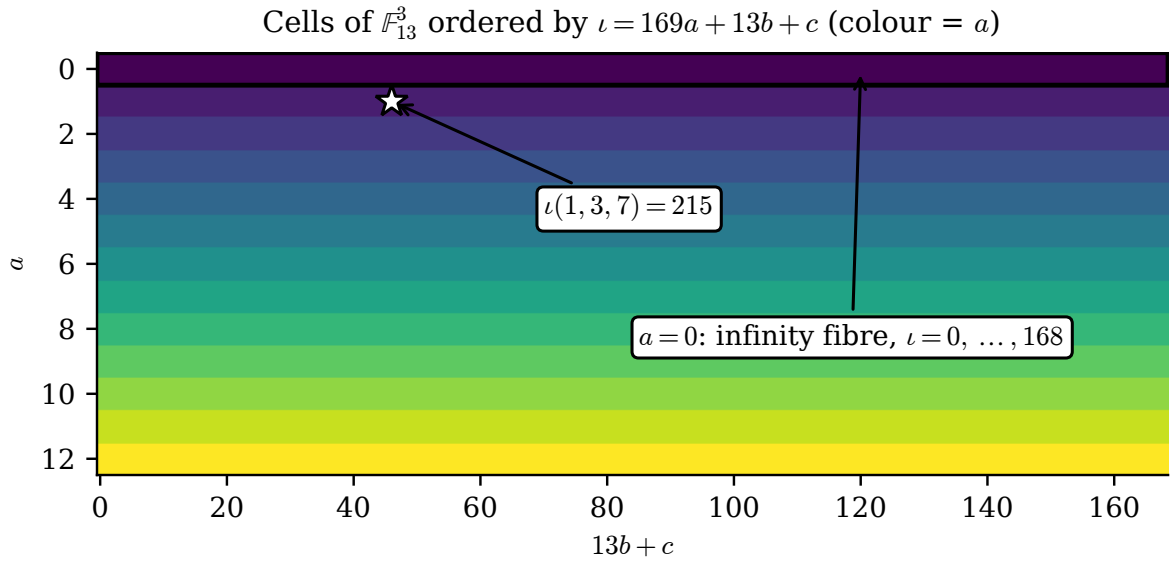


Figure 1: Cells of $\mathbb{F}_{13}^3$ in the order of the linear index, Eq. (9). Rows: leading digit a; columns: $13b + c$. The row $a = 0$ is the infinity fibre of Sect. 9 ($\iota = 0, \ldots, 168$); the star marks $\iota(1,3,7) = 215$.

Proposition 3.2. *The map* $\iota : \mathbb{F}_{13}^3 \to \{0, 1, \ldots, 2196\}$ *is a bijection, with inverse*

$$m \longmapsto \big(\lfloor m/169 \rfloor,\ \lfloor m/13 \rfloor \bmod 13,\ m \bmod 13\big).$$

Proof. Each integer $0 \le m < 13^3$ has a unique expansion $m = 169\alpha + 13\beta + \gamma$ with digits $\alpha, \beta, \gamma \in \{0, \ldots, 12\}$, and ι is this expansion read on the representatives $\bar{a}, \bar{b}, \bar{c}$. □

For example, $\iota(1,3,7) = 169 + 39 + 7 = 215$ and $\iota(12,12,12) = 2196$; the number of cells is $|\mathbb{F}_{13}^3| = 13^3 = 2197$ (Fig. 1).

Definition 3.3 (Coordinate mismatch)**.** For $u, w \in V$,

$$\mu(u, w) = \#\{\, i \in \{0,1,2\} : u_i \neq w_i \,\}. \tag{10}$$

The mismatch counts differing integer coordinates; no order on $\mathbb{F}_{13}$ and no notion of size of a field element is used.

Proposition 3.4. *μ is the Hamming distance on $\mathbb{F}_{13}^3$; it is a metric with values in $\{0,1,2,3\}$, and $\mu(u,w) = 0$ if and only if $u = w$. For fixed u, the number of w with $\mu(u,w) = d$ is $\binom{3}{d}12^d$, and these numbers sum to 13^3.*

Proof. Symmetry and the triangle inequality are those of the Hamming distance, and the range is bounded by the number of coordinates. To count the vectors at distance d, choose the d coordinates that differ ($\binom{3}{d}$ ways) and a different value in each (12 choices). The binomial theorem gives $\sum_d \binom{3}{d}12^d = (1+12)^3$. □

For instance, $\mu((1,2,3),(1,0,3)) = 1$. The distribution is shown in Fig. 2.

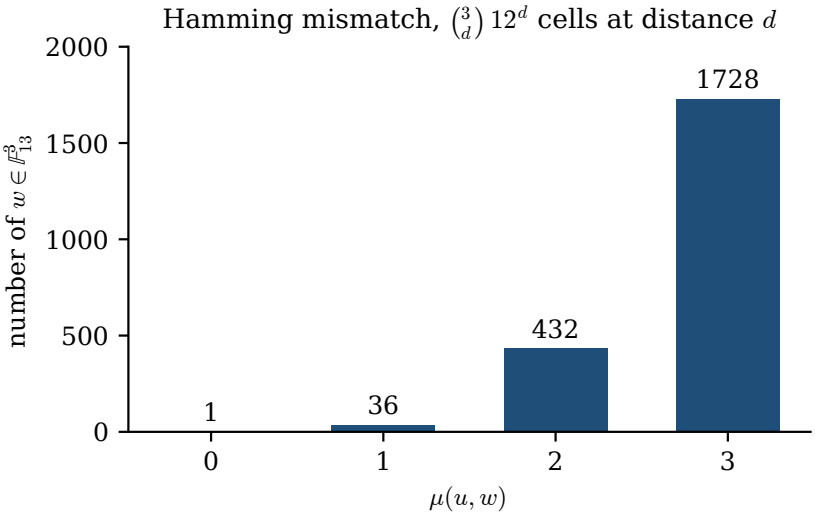


Figure 2: Distribution of $\mu(u, w)$, Eq. (10), over all 2197 vectors w for fixed u: the number of w at distance d is $\binom{3}{d}12^d = 1, 36, 432, 1728$ (Proposition 3.4).

Algorithm 1 `evolveFrom`(Ψ_b, s)

pert $\leftarrow \Psi_b \oplus s$
coupled $\leftarrow \mathcal{D}_f(\text{pert}; D_W)$
out $\leftarrow (\text{coupled}^+,\ \psi_b^-)$
$r \leftarrow P_{23}\,\text{out}^+$
$k \leftarrow 0$
if h is non-empty **then**
    Record(s, out) ▷ Algorithm 4
end if
return out

4 Evolution and the Golden Orbit

4.1 Evolution rule

Let Ψ_b be a base fermion and s a stimulus. The evolution rule is

$$\begin{aligned} \text{pert} &= \Psi_b \oplus s, \\ \text{coupled} &= \mathcal{D}_f(\text{pert}; D_W), \\ E(\Psi_b, s) &= \big(\text{coupled}^+,\ \psi_b^-\big). \end{aligned} \tag{11}$$

The plus component of the response is the coupled one, and the minus component of the base is retained. The response to a stimulus s is $E(\Psi_{\text{vac}}, s)$. Each application of E updates the state of the learner: it sets the residue $r \leftarrow P_{23}\,E(\Psi_b, s)^+$, resets the golden index $k \leftarrow 0$, and, if the history is non-empty, records an experience (Algorithm 4).

By Eq. (6), the response splits as $\text{out}^+ = a + r$ with $a = P_1\,\text{out}^+$ in the fixed sector and $r = P_{23}\,\text{out}^+$ the residue. In particular the vacuum has zero residue, since $P_{23}v_0 = 0$.

4.2 The golden group and the orbit of corrections

Definition 4.1 (Golden element)**.** Let $\Phi = 7 + 8\theta \in \mathbb{F}_{169}$.

Proposition 4.2. *(i)* $N(\Phi) = -1$ *and* $\sigma(\Phi) = \Phi^{13}$.

Algorithm 2 `evolveFromOrbit`(Ψ_b, s, j)

$\tilde{\Psi} \leftarrow \textsc{evolveFrom}(\Psi_b, s)$
$k \leftarrow \gamma(j)$
if $k = 0$ **or** $r = 0$ **then return** $\tilde{\Psi}$
end if
return $(B_k\tilde{\Psi}^+,\ \tilde{\Psi}^-)$

(ii) $\Phi^{14} = -1$ *and* Φ *has multiplicative order exactly* 28*; in particular the powers* $\Phi^0, \dots, \Phi^{27}$ *are pairwise distinct.*

(iii) The elements $g = 3 + 2\theta$ *and* $\rho = 8 + 5\theta$ *satisfy* $g = \Phi^{18}$, $\rho = \Phi^{26} = g^3$*; both have order* 14 *and lie in the subgroup* $\langle \Phi^2 \rangle$ *of order* 14*, which is the group of elements of norm one.*

Proof. (i) $N(\Phi) = 7^2 - 2 \cdot 8^2 = -79 \equiv -1 \pmod{13}$, and σ is the Frobenius map $x \mapsto x^{13}$. (ii) By (i), $\Phi^{14} = \Phi\,\sigma(\Phi) = N(\Phi) = -1$, so $\Phi^{28} = 1$ and the order divides 28 but not 14. The only divisors of 28 that do not divide 14 are 4 and 28, and $\Phi^2 = 8 + 8\theta$, $\Phi^4 = 10 + 11\theta \neq 1$; hence the order is 28. (iii) Direct computation in $\mathbb{F}_{169}$ gives $\Phi^{18} = 3 + 2\theta$ and $\Phi^{26} = 8 + 5\theta$, and $(3 + 2\theta)^3 = 8 + 5\theta$. Then $\gcd(18, 28) = \gcd(26, 28) = 2$ gives order 14, and $\langle \Phi^2 \rangle$ is the unique subgroup of order 14 of the cyclic group $\langle \Phi \rangle$, which coincides with the norm-one group because $N(x) = x^{14}$. □

The correction maps are indexed by this group.

Definition 4.3 (Correction maps and orbit step)**.** For $k \in \{0, \dots, 27\}$ let $B_k \in M_3(\mathbb{F}_{13})$ be the correction map attached to Φ^k (`goldenB`), with $B_0 = I$, and let $\gamma(j) \in \{0, \dots, 27\}$ denote the reduction of an integer index j (`goldenIndex`). For $\tilde{\Psi} = E(\Psi_b, s)$ with residue r and $k = \gamma(j)$ the orbit step is

$$E_j(\Psi_b, s) = \begin{cases} \tilde{\Psi} & \text{if } k = 0 \text{ or } r = 0, \\ \left(B_k\tilde{\Psi}^+,\ \tilde{\Psi}^-\right) & \text{otherwise,} \end{cases} \tag{12}$$

and the golden index of the learner is set to k. The *orbit* of (Ψ_b, s) is the list $\left(B_j\tilde{\Psi}^+, \tilde{\Psi}^-\right)_{j=0}^{27}$ if $r \neq 0$, and the singleton $\{\tilde{\Psi}\}$ if $r = 0$.

Proposition 4.4. *If the residue of the response vanishes, every* E_j *equals* $\tilde{\Psi}$ *and the orbit is a singleton. In particular the orbit of the vacuum has one element.*

Proof. Immediate from Eq. (12) and the definition of the orbit; the vacuum has zero residue because $P_{23}v_0 = 0$. □

5 Selection of a Correction

5.1 Selection for a single trial

Given a base Ψ_b, a stimulus s and a target $w \in V$, the routine `pickOrbit` evaluates $\tilde{\Psi} = E(\Psi_b, s)$ and scans $j = 0, \dots, 27$. Write $\mu_j = \mu(B_j\tilde{\Psi}^+, w)$ and define the *ledger score*

$$\mathrm{sc}(j) = \begin{cases} \max\limits_{\xi \in L,\ \xi.j = j} \mathrm{matchDepth}(h, \xi.\delta) & \text{if } \exists \xi \in L :\ \xi.j = j, \\ -1 & \text{otherwise (in particular if } L = \emptyset). \end{cases} \tag{13}$$

Here matchDepth is the prefix depth of Sect. 7 and $\xi.j$, $\xi.\delta$ are fields of ledger records (Sect. 8).

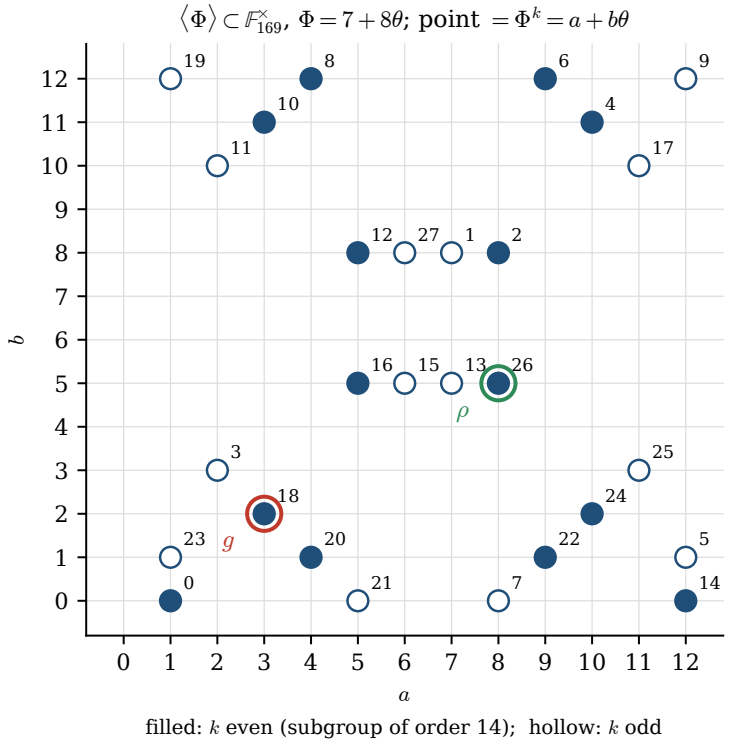


Figure 3: The cyclic group $\langle\Phi\rangle \subset \mathbb{F}_{169}^{\times}$, $\Phi = 7 + 8\theta$ (Proposition 4.2). A point is $\Phi^k = a + b\theta$ in the 13×13 grid of $\mathbb{F}_{169}$; filled points (k even) form the subgroup of order 14 containing g ($k = 18$) and ρ ($k = 26$).

Algorithm 3 `pickOrbit`(Ψ_b, s, w)

$\tilde{\Psi} \leftarrow$ EvolveFrom(Ψ_b, s)
if $r = 0$ **then**
  $k \leftarrow 0$; **return** $(0,\ \tilde{\Psi},\ \mu(\tilde{\Psi}^+, w),\ 1)$
end if
$(j^*, e^*, c^*) \leftarrow (0,\ 4,\ -1)$; seen $\leftarrow \emptyset$
for $j = 0, \ldots, 27$ **do**
  $p \leftarrow B_j\tilde{\Psi}^+$; add p to seen; $e \leftarrow \mu(p, w)$
  $c \leftarrow \mathrm{sc}(j)$ if $e \le e^*$, else -1
  if $e < e^*$ **or** $(e = e^*$ **and** $c > c^*)$ **then**
    $(j^*, e^*, c^*) \leftarrow (j, e, c)$; best $\leftarrow (p, \tilde{\Psi}^-)$
  end if
end for
$k \leftarrow j^*$; **return** $(j^*,\ \text{best},\ e^*,\ |\text{seen}|)$

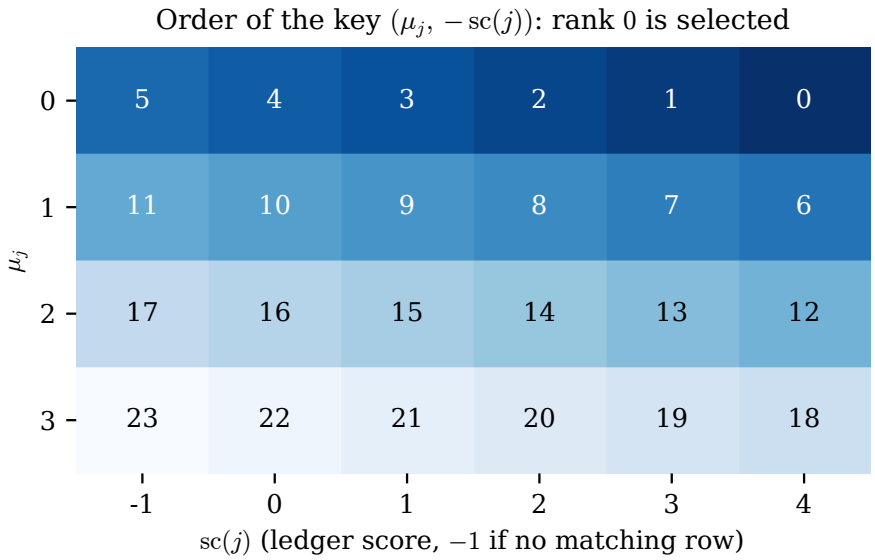


Figure 4: Total order on the keys $(\mu_j, -\mathrm{sc}(j))$ used by `pickOrbit` (Proposition 5.1, Eq. (13)). Rank 0 is selected; a smaller mismatch μ_j always beats a larger ledger score, and the score only breaks ties.

Proposition 5.1 (Selection rule). *Assume $r \neq 0$. Then the index returned by `pickOrbit` is the smallest $j \in \{0, \dots, 27\}$ that minimises the pair $(\mu_j, -\mathrm{sc}(j))$ in lexicographic order. The returned error lies in $\{0, 1, 2, 3\}$. If the ledger is empty, then $\mathrm{sc} \equiv -1$ and the returned index is the smallest j minimising μ_j.*

Proof. Since $\mu_j \leq 3 < 4$, the first iteration always satisfies $e < e^*$ and installs $(0, \mu_0, \mathrm{sc}(0))$. Suppose that after iteration $j-1$ the triple holds the lexicographic minimum of $(\mu_i, -\mathrm{sc}(i))$ over $i < j$, attained first at j^*. At iteration j the score $\mathrm{sc}(j)$ is used exactly when $\mu_j \leq e^*$, and the update fires if and only if $\mu_j < e^*$ or $(\mu_j = e^*$ and $\mathrm{sc}(j) > c^*)$, that is, if and only if $(\mu_j, -\mathrm{sc}(j))$ is strictly smaller than the current pair. The strict comparison keeps the earlier index on ties, so the invariant persists. The ledger is not modified during the scan (its only write occurs inside the preceding call of `evolveFrom`). If $L = \emptyset$ then $\mathrm{sc}(j) = -1$ for all j. □

5.2 Selection over a sequence of trials

Given trials $\mathcal{T} = \{(\Psi_b^{(t)}, s^{(t)}, w^{(t)})\}_{t=1}^n$, the routine `fitOrbit` looks for one stationary index for the whole labelled sequence:

$$J(j) = \sum_{t=1}^{n} \mu\big(E_j(\Psi_b^{(t)}, s^{(t)})^+,\, w^{(t)}\big), \qquad j^* = \min \arg\min_{0 \leq j < 28} J(j). \tag{14}$$

The initial bound is $4n+1 > 3n \geq J(j)$ and the comparison is strict, so that ties keep the smallest index (the original, uncorrected transform when 0 is among the minimisers). The returned state is the state of the last trial under j^*, and the returned error is $J(j^*) \in \{0, \dots, 3n\}$. For $\mathcal{T} = \emptyset$ the result is $(0, \Psi_{\mathrm{vac}}, 0, 1)$. Each of the $28n$ evaluations calls `evolveFromOrbit`, hence `evolveFrom` once (Algorithm 1).

6 Relaxations and Chirality

6.1 Quadratic relaxation

For a fermion Ψ let $n_+ = n_+(\Psi)$ and $n_- = n_-(\Psi)$; here n_+ is a homogeneous quadratic form in ψ^+. The routine `relaxQuadratic` rescales the plus component so that the two quadratic

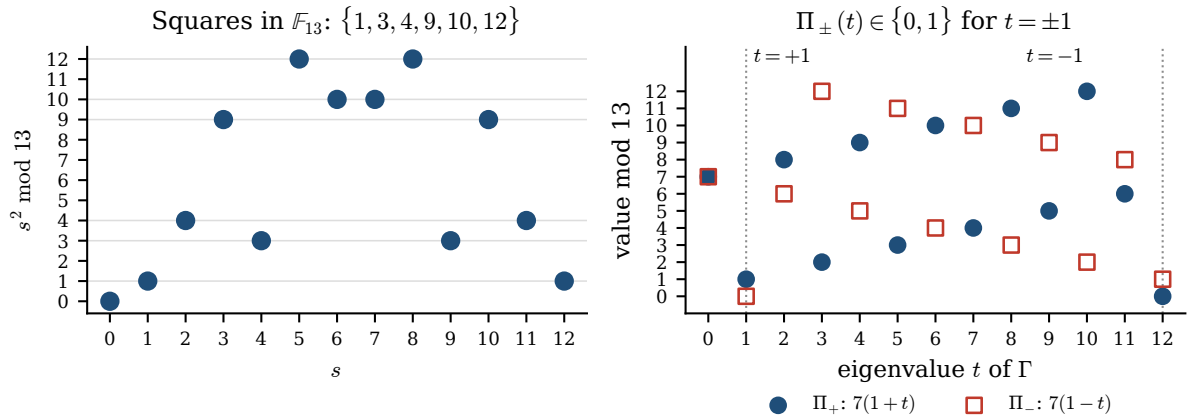


Figure 5: Left: the squares of $\mathbb{F}_{13}$ (quadratic relaxation). Right: the projectors $\Pi_\pm = 7(I \pm \Gamma)$ of Eq. (16) on the eigenvalues $t = \pm 1$ of Γ: $\Pi_+(1) = 1$, $\Pi_+(-1) = 0$, $\Pi_-(1) = 0$, $\Pi_-(-1) = 1$ (Proposition 6.2).

invariants agree, whenever this is possible in $\mathbb{F}_{13}$:

1. if $n_+ = n_-$: return Ψ unchanged (scale 1, target 1);
2. if $n_+ = 0$ and $n_+ \neq n_-$: return Ψ unchanged (scale 1, target 0), no solution;
3. otherwise let $\tau_q = n_- \, n_+^{-1} \in \mathbb{F}_{13}$. If τ_q is a square with root $s = \texttt{sqrtF13}(\tau_q)$, return $(s\,\psi^+, \psi^-)$ with scale s; otherwise return Ψ unchanged, no solution.

The nonzero squares of $\mathbb{F}_{13}$ are $\{1, 3, 4, 9, 10, 12\}$ (Fig. 5, left).

Proposition 6.1. *In case 3, when a root s exists, $n_+(s\,\psi^+) = n_-$.*

Proof. Homogeneity of degree two gives $n_+(s\psi^+) = s^2 n_+(\psi^+) = \tau_q \, n_+ = n_-$. □

6.2 Projection to the fixed sector

The routine `relaxP1` sets $r \leftarrow P_{23}\psi^+$ and returns $(P_1\psi^+, \psi^-)$; the residue is stored, not discarded. By Eq. (6), $P_1\psi^+ + r = \psi^+$.

6.3 Chiral decomposition

Since $2 \cdot 7 = 14 \equiv 1$ in $\mathbb{F}_{13}$, we have $2^{-1} = 7$. The chirality operator γ_5 acts on $\mathbb{F}_{13}^6 = V \times V$ as

$$\Gamma = \varepsilon \otimes G, \tag{15}$$

where G is the matrix of Eq. (7) and $\varepsilon \in M_2(\{0, \pm 1\})$ is a sign convention with $\varepsilon^2 = I_2$ and $\operatorname{tr} \varepsilon = 0$ (for example $\operatorname{diag}(1, -1)$ or $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$). The chiral halves of a fermion Ψ are

$$\Pi_\pm = 7\,(I \pm \Gamma), \qquad \Psi_\pm = \Pi_\pm \Psi = \big(7(\psi^+ \pm \Gamma\psi^+),\ 7(\psi^- \pm \Gamma\psi^-)\big). \tag{16}$$

Proposition 6.2 (Chiral decomposition)**.** *$\Gamma^2 = I$, $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+\Pi_- = 0$, $\Pi_+ + \Pi_- = I$ and $\Gamma\Pi_\pm = \pm\Pi_\pm$. Consequently $\mathbb{F}_{13}^6 = \operatorname{im}\Pi_+ \oplus \operatorname{im}\Pi_-$, and both summands have dimension three.*

Proof. By the mixed-product property, $\Gamma^2 = \varepsilon^2 \otimes G^2 = I_2 \otimes I_3 = I_6$, using $G^2 = I$. Next $(I \pm \Gamma)^2 = I \pm 2\Gamma + \Gamma^2 = 2(I \pm \Gamma)$, so $\Pi_\pm^2 = 49 \cdot 2\,(I \pm \Gamma) = 98\,(I \pm \Gamma)$, and $98 \equiv 7 \pmod{13}$; thus $\Pi_\pm^2 = \Pi_\pm$. Further $\Pi_+\Pi_- = 49\,(I - \Gamma^2) = 0$, $\Pi_+ + \Pi_- = 14\,I = I$, and $\Gamma\Pi_\pm = 7(\Gamma \pm \Gamma^2) = 7(\Gamma \pm I) = \pm\Pi_\pm$. Since $\Gamma^2 = I$ and 2 is invertible, Γ is diagonalisable with eigenvalues ± 1; let $m_\pm$ be their multiplicities, so $m_+ + m_- = 6$ and $\operatorname{tr}\Gamma \equiv m_+ - m_- \pmod{13}$. But $\operatorname{tr}\Gamma = \operatorname{tr}\varepsilon \cdot \operatorname{tr} G = 0$. As $|m_+ - m_-| \leq 6 < 13$, we get $m_+ = m_- = 3$, and $\operatorname{im}\Pi_\pm$ is the (± 1)-eigenspace of Γ. □

7 The Witt History

A *Witt prefix* is a finite digit string $h = (d_0, \dots, d_{n-1})$ with $d_r \in \{0, \dots, 12\}$ (class `WittPrefix`). It represents the truncation of a 13-adic integer (see [2]):

$$x(h) = \sum_{r=0}^{n-1} d_r \, 13^r \in \mathbb{Z}, \qquad 0 \le x(h) < 13^n, \tag{17}$$

computed exactly with arbitrary-precision integers. The operation append(a) adds the digit $\bar{a}$ at position n, so that x grows by $\bar{a} \cdot 13^n$. The routine `rememberDigits` appends a list of digits, and `rememberStim`(s) appends the three digits of s^-; each stimulus therefore increases the depth by three. Depth is history: the position of a digit records when it was learned, and a later symbol never overwrites an earlier one.

Two comparison functions are used:

$$\text{matchDepth}(h, h') = \text{number of common initial digits},$$
$$\text{tailMatch}(h, h') = \text{number of common final digits}.$$

Proposition 7.1. *For digit strings h, h' of lengths n, n',*

$$\text{matchDepth}(h, h') = \max\{\, d \le \min(n, n') : \; x(h) \equiv x(h') \pmod{13^d} \,\},$$

that is, the prefix depth is the 13-adic closeness of the two truncations.

Proof. Because $0 \le d_r \le 12$, the residue $x(h) \bmod 13^d$ equals $\sum_{r<d} d_r 13^r$ and determines the first d digits uniquely. Hence $x(h) \equiv x(h') \pmod{13^d}$ if and only if the first d digits agree. □

Proposition 7.2. *If the history is empty and a single stimulus $s = (\cdot, (a, b, c))$ is remembered, then $x = a + 13b + 169c = \iota(c, b, a)$. Thus the Witt value of a single learned cell is the linear index (9) of the digit-reversed triple.*

Proof. The digits are appended in the order a, b, c, so $d_0 = \bar{a}$, $d_1 = \bar{b}$, $d_2 = \bar{c}$, and Eq. (17) gives $\bar{a} + 13\bar{b} + 169\bar{c} = \iota(c, b, a)$. □

For example, remembering $s^- = (1, 0, 2)$ gives depth 3 and $x = 1 + 0 \cdot 13 + 2 \cdot 169 = 339$; the digits $[0, 4, 7]$ (first digit 0) give $x = 4 \cdot 13 + 7 \cdot 169 = 1235$.

8 Experience Ledger and Associative Retrieval

8.1 The ledger

Definition 8.1 (Experience record)**.** An *experience* is a tuple $\xi = (\delta, \varphi, j, \mathbf{s}, \mathbf{p}, n)$, where δ is a digit string (a copy of the history at recording time), φ is the integer identifier of the spectral flag of the learned state, j is the golden index at recording time, $\mathbf{s} \in \{0, \dots, 12\}^3$ is the stimulus (the minus component of the stimulus fermion), $\mathbf{p}$ is the learned plus component, and $n \in \mathbb{Z}_{\ge 1}$ is a counter. Counters are integers and are never reduced modulo 13.

The ledger has capacity

$$\text{CAP} = 13^3 = 2197, \tag{18}$$

the size of the stimulus census of Sect. 3.

Proposition 8.2. *After every call of* Record, $|L| \le \text{CAP}$. *A record already present is never duplicated; its counter is incremented by 1 and its plus component is overwritten.*

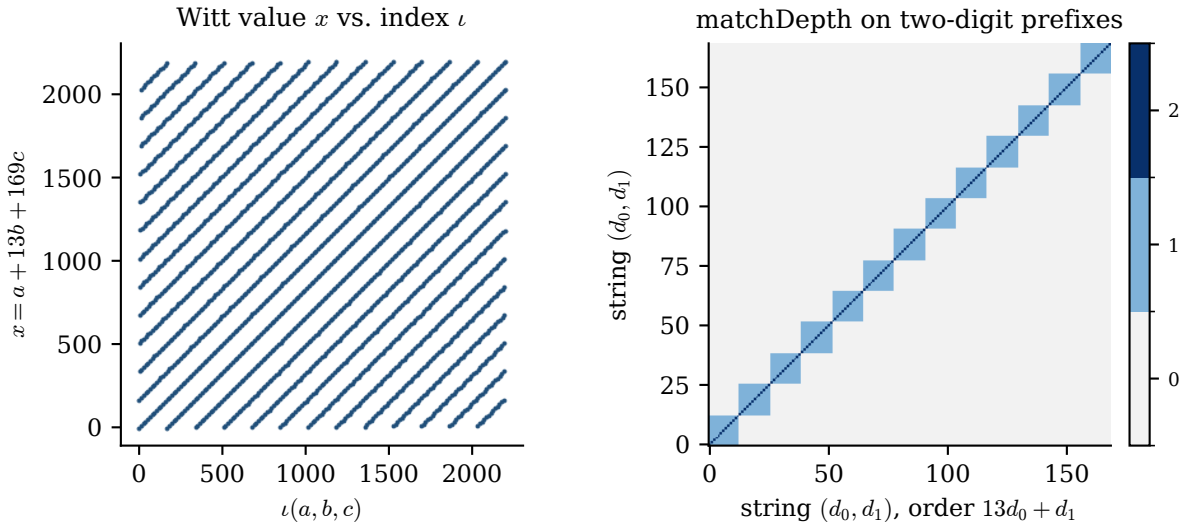


Figure 6: Left: the Witt value $x = a + 13b + 169c$, Eq. (17), against the index $\iota(a, b, c)$, Eq. (9); the picture is the digit reversal of Proposition 7.2. Right: `matchDepth` on the 169 two-digit prefixes (Proposition 7.1): depth 2 on the diagonal, depth 1 inside the 13×13 blocks with common first digit, depth 0 elsewhere.

Algorithm 4 Record$(s, \Psi_{\text{learned}})$ (`rememberExperience`)

$\varphi \leftarrow$ `spectralFlag`$(\Psi^{+}_{\text{learned}}, R)$.flagId; $\mathbf{s} \leftarrow s^{-}$; $\mathbf{p} \leftarrow \Psi^{+}_{\text{learned}}$; $\delta \leftarrow$ copy of h
if $\exists \xi \in L$ with $(\xi.\varphi, \xi.j, \xi.\mathbf{s}, \xi.\delta) = (\varphi, k, \mathbf{s}, \delta)$ **then**
    $\xi.n \leftarrow \xi.n + 1$; $\xi.\mathbf{p} \leftarrow \mathbf{p}$; **return** ξ
end if
append $(\delta, \varphi, k, \mathbf{s}, \mathbf{p}, 1)$ to L
if $|L| >$ CAP **then** remove the $|L| -$ CAP oldest records
end if
return the new record

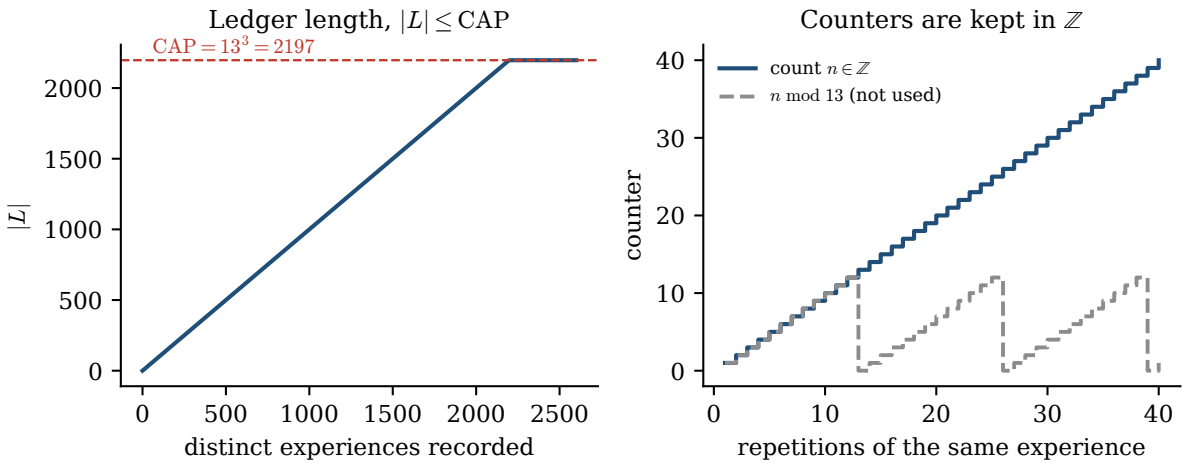


Figure 7: Left: ledger length against the number of distinct experiences recorded, $|L| \le \text{CAP} = 13^3$ (Eq. (18), Proposition 8.2). Right: the counter of a repeated experience is kept in $\mathbb{Z}$ and never reduced mod 13.

Proof. A call appends at most one record, and if the length then exceeds CAP the oldest surplus records are removed. The search for an existing record precedes the append. □

Remark 8.3. Records are written by `evolveFrom`, only when the history is non-empty, and Algorithm 1 resets $k \leftarrow 0$ immediately before the call. The golden index stored in a record is the value of k at the moment of writing.

8.2 Spectral flag and the invariant τ

The memory keys of an experience are derived from a spectral invariant of the learned plus component. For $z \in V$ let

$$\tau(z) = (Q_2 z)_i \left((Q_3 z)_i\right)^{-1} \in \mathbb{F}_{169}, \tag{19}$$

where i is the first index with $(Q_3 z)_i \neq 0$, and Q_2, Q_3 are the spectral projectors of Sect. 2.

Proposition 8.4 (Properties of τ)**.** *Let $z \in V$.*

(i) *$\tau(z)$ is defined if and only if $r(z) = P_{23} z \neq 0$. Hence it is undefined exactly on the 13 vectors of the fixed sector, in particular on the vacuum vector v_0, and defined on the remaining 2184 vectors.*

(ii) *Where defined, $\tau(z) \neq 0$ and $N(\tau(z)) = 1$.*

(iii) *$\tau(cz) = \tau(z)$ for $c \in \mathbb{F}_{13}^{\times}$.*

(iv) *$\tau(T_3 z) = \rho\, \tau(z)$, with ρ of order 14 as in Eq. (4).*

(v) *The image of τ is the whole norm-one group $\langle \Phi^2 \rangle$ of order 14.*

Proof. (i) Since $Q_2 P_1 = Q_3 P_1 = 0$, we have $Q_i z = Q_i P_{23} z$; hence $P_{23} z = 0$ implies $Q_2 z = Q_3 z = 0$. Conversely, z has entries in $\mathbb{F}_{13}$, so $Q_2 z = \sigma(Q_3 z)$; thus $Q_3 z = 0$ implies $Q_2 z = 0$ and $P_{23} z = Q_2 z + Q_3 z = 0$. The fixed sector $\operatorname{im} P_1$ has dimension one and 13 elements. (ii) $(Q_2 z)_i = \sigma\big((Q_3 z)_i\big) \neq 0$, so $\sigma(\tau(z)) = \tau(z)^{-1}$ and $N(\tau(z)) = \tau(z)\,\sigma(\tau(z)) = 1$. (iii) The scalar c cancels in the ratio. (iv) $Q_i T_3 = \lambda_i Q_i$ gives $Q_2 T_3 z = \lambda_2 Q_2 z$ and $Q_3 T_3 z = \lambda_3 Q_3 z$; as $\lambda_3 \neq 0$, the index i is unchanged and $\tau(T_3 z) = (\lambda_2/\lambda_3)\,\tau(z)$. The order of ρ is 14 by Proposition 4.2. (v) By (ii) the image lies in the norm-one group, which has 14 elements and equals $\langle \Phi^2 \rangle$; exhaustive evaluation of τ over the 2184 vectors of (i) attains all 14 values. □

The *spectral flag* of a vector z is the discretisation of $\tau(z)$ into an integer identifier $\varphi \in \{0, \dots, \text{FLAG_COUNT} - 1\}$ (`spectralFlag`), and the *panel* $\mathcal{N}(\varphi)$ of a flag is the set of its

neighbours (`panelNeighbors`); flags with a negative identifier take part in neither the flag nor the panel stage of the retrieval below. By Proposition 8.4(iv), the register T_3 acts on the values of τ by multiplication by ρ, which links the memory keys to the dynamics of the register.

8.3 Associative retrieval

Let $v = \Psi^+$ be a plus component, let φ_0 be the identifier of its spectral flag, and let h be the current history. For each $\xi \in L$ put

$$\mathrm{pre}(\xi) = \mathrm{matchDepth}(h, \xi.\delta), \qquad \mathrm{tail}(\xi) = \mathrm{tailMatch}(h, \xi.\delta).$$

Set $\mathcal{N} = \mathcal{N}(\varphi_0)$ if $\varphi_0 \geq 0$ and $\mathcal{N} = \emptyset$ otherwise. The result of `searchMemory` consists of three lists:

$$\begin{aligned} \text{context} &= \big(\xi \in L : \ \mathrm{pre}(\xi) > 0 \text{ or } \mathrm{tail}(\xi) > 0\big), \text{ sorted by } (\mathrm{pre}, \mathrm{tail}, \xi.n) \text{ descending}, \\ \text{flag} &= \{\, \xi \in L : \ \varphi_0 \geq 0, \ \xi.\varphi = \varphi_0 \,\}, \\ \text{panel} &= \{\, \xi \in L : \ \xi.\varphi \in \mathcal{N} \,\}. \end{aligned} \tag{20}$$

The sort is lexicographic (prefix depth first, then tail match, then counter) and stable, so ties keep the order of the ledger; the lists *flag* and *panel* keep ledger order. The retrieval order is thus: Witt prefix, then flag, then panel, then golden index through the score sc of Eq. (13). The retrieval complements, and does not replace, the selection of Sect. 5 and the ratio search of Sect. 9. The formatted summary of a result has the form "Memory: prefix p flag f panel q ledger $\ell/2197$", where p, f, q are the lengths of the three lists and $\ell = |L|$.

9 The Stimulus Index

The routine `StimulusIndex.build` enumerates all cells $(a, b, c) \in \mathbb{F}_{13}^3$ in lexicographic order. For each cell it forms the elementary stimulus $s = (0, (a, b, c))$, evolves it from the vacuum, and stores the row

$$\begin{aligned} \tilde{\Psi} &= E(\Psi_{\mathrm{vac}}, s), \quad n_\pm = n_\pm(\tilde{\Psi}), \quad \text{ratio} = n_-/n_+ \ \ (\text{or } -1 \text{ if } n_+ = 0), \\ \text{phase} &= \mathrm{phase}_{168}(\tilde{\Psi}), \quad \epsilon = g_{00}(\tilde{\Psi}) \bmod 2, \quad (\text{face}, \text{comp}) = \texttt{fermionToPlaquette}(\text{phase}, \epsilon), \\ \text{index} &= \iota(a, b, c), \quad q = q_{\mathrm{top}}(\tilde{\Psi}). \end{aligned}$$

Rows are also grouped by the value n_+. The lookups are `at`(a, b, c), the row at position $\iota(a, b, c)$; `byMinus`(v), the same for $v \in V$; and `firstWithNplus`(n), the first row with $n_+ = n \bmod 13$. The index is built once and cached.

Proposition 9.1. *The row list has* 2197 *entries, and* `at`(a, b, c) *returns the row whose* `index` *field is* $\iota(a, b, c)$.

Proof. Lexicographic order of (a, b, c) is increasing order of ι by Eq. (9); by Proposition 3.2 the positions $0, \ldots, 2196$ are each hit exactly once. □

First square ratio. The routine `firstSquareRatio`(requireQ) scans the rows in order and discards a row if $n_+ = 0$, if $n_+ = n_-$, or if the ratio is undefined. Among the remaining rows it considers those whose ratio is a square in $\mathbb{F}_{13}$. If requireQ is true it returns the first such row with $q \neq 0$, that is, a state with non-trivial topological charge q_{top}; if no such row exists, or if requireQ is false, it returns the first row with a square ratio.

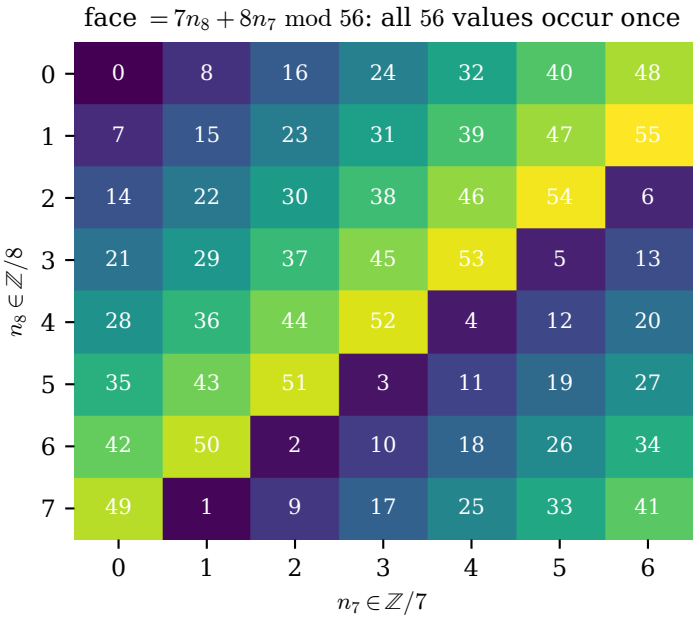


Figure 8: The face map of Eq. (21): $(n_8, n_7) \mapsto 7n_8 + 8n_7 \bmod 56$ takes each value $0, \ldots, 55$ exactly once (Proposition 9.3).

Infinity fibre. The routine `infinityFiber` returns the rows with $a = 0$; there are $13 \times 13 = 169$ of them.

Proposition 9.2. *The infinity fibre consists of the rows at positions* $0, 1, \ldots, 168$.

Proof. For $a = 0$ the index is $\iota = 13\bar{b} + \bar{c}$, which ranges over $0, \ldots, 168$ as (b, c) ranges over $\mathbb{F}_{13}^2$, by Proposition 3.2 restricted to numbers with leading digit 0. □

9.1 Plaquette labels

The plaquette map $(n, \epsilon) \mapsto (\text{face}, \text{comp})$, with $n \in \mathbb{Z}/168$ and $\epsilon \in \{0, 1\}$, is a bijection onto $[0, 56) \times [0, 6)$, with inverse `plaquetteToFermion`; both sets have $336 = 168 \cdot 2 = 56 \cdot 6$ elements. The face label is obtained from the Chinese-remainder components $n_8 = n \bmod 8$ and $n_7 = n \bmod 7$ by

$$(n_8, n_7) \longmapsto (7n_8 + 8n_7) \bmod 56. \tag{21}$$

Proposition 9.3. *The map* (21) *is a bijection* $\mathbb{Z}/8 \times \mathbb{Z}/7 \to \mathbb{Z}/56$.

Proof. Modulo 8 the value is $7n_8 \equiv -n_8$, and modulo 7 it is $8n_7 \equiv n_7$. By the Chinese remainder theorem $\mathbb{Z}/56 \cong \mathbb{Z}/8 \times \mathbb{Z}/7$, so the map corresponds to $(n_8, n_7) \mapsto (-n_8, n_7)$, which is a bijection. □

10 Finiteness and Computational Cost

All state of the learner is finite:

- a fermion lives in $\mathbb{F}_{13}^6$, so there are 13^6 states;
- the ledger holds at most 2197 records (Proposition 8.2);
- the stimulus index has 2197 rows (Proposition 9.1);
- an orbit has at most 28 elements.

The number of evolution calls performed by each routine follows from Algorithms 1 to 3 and Eq. (14):

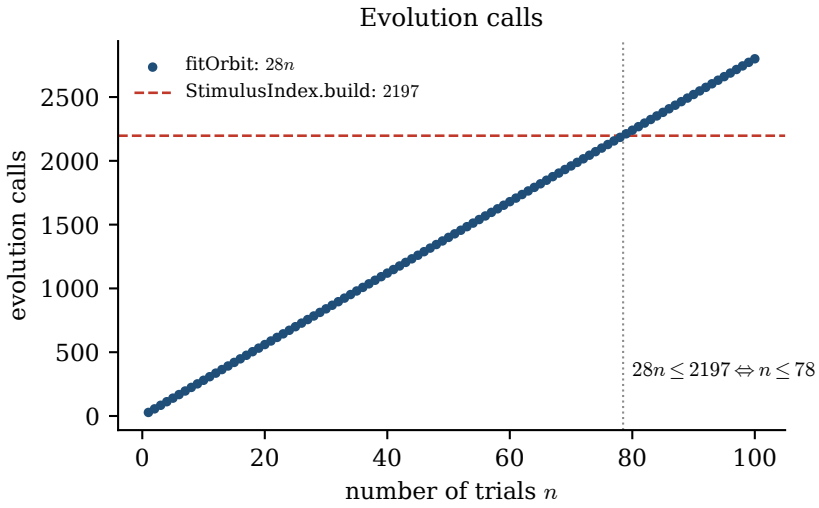


Figure 9: Evolution calls of `fitOrbit` ($28n$ for n trials, Eq. (14)) compared with the fixed 2197 calls of `StimulusIndex.build`: $28n \le 2197$ if and only if $n \le 78$.

Routine	Evolution calls
`StimulusIndex.build`	2197
`pickOrbit`	1 evolution, 28 candidate maps B_j
`fitOrbit` on n trials	$28n$
`orbitOf`	1 evolution, 28 candidate maps B_j

A call of `searchMemory` scans the ledger a constant number of times, so it examines at most $3 \cdot 2197$ records, each compared with the history as a digit string.

11 Conclusion

We have specified the learning mechanism of the Zeta project as a finite state machine over $\mathbb{F}_{13}$ and $\mathbb{F}_{169}$: an exact evolution rule, a cyclic family of 28 correction maps indexed by the subgroup $\langle \Phi \rangle$ of $\mathbb{F}_{169}^{\times}$, a selection rule that minimises an integer mismatch and breaks ties by memory, a 13-adic history in which prefix depth measures closeness, and a bounded integer ledger with a three-stage associative search keyed by the spectral invariant τ. The properties established in the text are elementary but load-bearing: the linear index is a bijection, the selection rule is a lexicographic minimum, the chiral halves are complementary subspaces of dimension three, prefix matching is 13-adic congruence, the ledger never exceeds 13^3 records, and τ is a scale-invariant, register-equivariant map into the norm-one group.

Because every quantity is a residue or an integer, all statements are exactly reproducible. The figures of this article are generated by the script `figures.py` distributed with the source, and the self-test suite listed in Appendix A exercises the properties on concrete instances.

A Self-Test Properties

The learner module includes a self-test routine (`learnerAudit`) that asserts the properties in Table 1. They are instances of the general statements above; the orbit properties refer to the stimulus $s_A = \big((2,3,1),(1,0,2)\big)$ evolved from the vacuum.

Table 1: Properties asserted by the self-test routine.

Property	Assertion
Vacuum and evolution	
Vacuum	$\Psi^+ = \Psi^- = (1,3,7)$
Evolution method	`evolve`$(\Psi_{\text{vac}}, s) =$ `evolveFrom`(Ψ_{vac}, s)
Minus component kept	the minus part of the output equals that of the base
Index 0	`evolveFromOrbit`$(\Psi_{\text{vac}}, s, 0) =$ `evolveFrom`(Ψ_{vac}, s)
Indexing and census	
Index of $(1,3,7)$	$\iota(1,3,7) = 215$
Maximal index	$\iota(12,12,12) = 2196$
Number of cells	2197 rows
Infinity fibre	$a = 0$ gives 169 cells
Row lookup	`at`$(1,3,7)$.index $= 215 =$ `rows[215].index`
Plaquettes	
Plaquette bijection	round trip for all $n < 168$, $\epsilon < 2$; face < 56, comp < 6
Face labels	the pairs (n_8, n_7) give 56 distinct faces
Learned row	round trip on the row of $(1,0,2)$
Chirality	
Reconstruction	$\Psi_+ + \Psi_- = \Psi$ on the vacuum
Chirality of Ψ_+	$\gamma_5 \Psi_+ = \Psi_+$ on the vacuum
Witt history	
Depth	one stimulus adds 3 digits
Witt value	$x = 339$ for $s^- = (1,0,2)$
Boundary case	digits $[0,4,7]$ give $x = 1235$
Sectors	
Vacuum sectors	$r = 0$ and a equals the vacuum plus part
Vacuum orbit	`orbitOf`$(\Psi_{\text{vac}}, 0)$ has 1 element
Residue of s_A	$r \neq 0$ and $a + r$ equals the learned plus part
Projection	`relaxP1`: residue $= r$, plus part $= a$
Reconstruction	$a + r =$ learned plus part
Orbit	
Orbit size for s_A	28 elements, with 28 distinct plus parts
Minus part	all minus parts equal the vacuum minus part
Fixed sector	P_1(plus part)$= a$ for every element
`pickOrbit`	target $B_5\tilde{\Psi}^+$: $j = 5$, error 0, 28 distinct images
`fitOrbit`	two trials: $j = 5$, error 0
Mismatch	$\mu((1,2,3),(1,0,3)) = 1$
Ledger and memory	
Counters in $\mathbb{Z}$	one record with count 1, then count 2 after a repetition
Live flag	flag of the learned state with $0 \leq \varphi <$ FLAG_COUNT
Search	at least one record in the flag list and in the context list
Capacity	CAP $= 2197 = 13^3$
Empty ledger	sc$(5) = -1$ for a new learner
Panel	$\lvert\mathcal{N}(\varphi)\rvert = 26$ for a live flag
`pickOrbit` with ledger	$j = 5$, error 0
Summary text	contains "prefix" and "2197"

References

[1] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Encyclopedia of Mathematics and its Applications, vol. 20, Cambridge University Press, 1997.

[2] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics, vol. 67, Springer, 1979.

[3] D. Navrátil, `bridge.ts`: exact arithmetic layer of the Zeta project (source module); defines `WittPrefix` and the spectral projectors used in Sect. 2.

[4] D. Navrátil, `learner.ts`: learning module of the Zeta project (source module).